Statistical Analysis and Reproducibility Appendix: PMData Development and Locked External Evaluation

Bounded tail-sensitive HRR distribution index

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S1. Purpose and analysis hierarchy

This appendix records the executable analysis specification and reviewer-requested robustness analyses. The participant was the unit of independence. Sessions and protocol stages were repeated observations. Reviewer-round-5 analyses directly compared mean HRR with mean HRR plus Δtilt, aligned inner selection with participant-balanced MAE, added transparent distribution baselines, and used repeated-measures agreement methods.

Supplementary Table S1. Analysis hierarchy

Level	Analysis	Interpretation
Principal in revised report	Participant-balanced outer-fold MAE: binned mean HRR versus binned mean HRR plus Δtilt; λ selected within each training fold	Exploratory development-sample comparison; not preregistered
Secondary	Within-person Spearman association; comparison with intercept-only model	Ranking and absolute calibration
Exploratory	Alternative formula families, load correlations, high-RPE AUC, ordinal agreement, matching and anchor sensitivities	Hypothesis-generating only

The principal comparison was designated during revision. No timestamped analysis plan, confirmatory p values, or multiplicity-adjusted superiority claims were available.

S2. Session reconstruction and quality control

PMSys reports were converted to Europe/Oslo local time and linked one to one to Fitbit tracker sessions with compatible activity labels. Candidate reports could occur from 15 minutes before to 12 hours after the tracker session. Duration difference could not exceed max(10 minutes, 35% of reported duration). A greedy assignment minimized report delay and proportional duration difference. The primary set further required a bidirectionally unique match, an RPE delay of 15-180 minutes, valid RPE, at least 80% valid heart-rate coverage, and at least 10 valid minutes. Heart-rate observations required Fitbit confidence at least 2 and 30-220 beats·min^{-1}. Each observation was weighted by the following interval. Intervals were capped at 30 seconds so missing gaps did not contribute observed time. The absence of a shared PMSys-Fitbit session identifier means the reconstructed match is not a verified gold standard.

Supplementary Table S2. Session-matching stability

Rule	Selected pairs	Bidirectionally unique	Primary pairs retained
Primary 15-180-min window	449	443	255 (100.0%)
15-90-min window	401	398	225 (88.2%)
30-120-min window	413	408	232 (91.0%)
Primary window + duration difference ≤10 min	363	357	214 (83.9%)
Delay-priority matching cost	449	443	255 (100.0%)
Duration-priority matching cost	449	443	255 (100.0%)

The retained-primary column tracks IDs from the original 255-session primary set. The rebuilt-analysis count is reported separately in Supplementary Table S27 because changing a matching rule can admit new session-report pairs after complete quality control. These quantities answer different questions and should not be interpreted as inconsistent denominators.

S3. Physiological anchors and formulas

$$HRR(t) = clip\{[HR(t) - HRrest] / [HRmax - HRrest], 0, 1\}.$$

HRrest was the participant median of Fitbit daily resting-heart-rate values, with a sleep-score value used when daily values were unavailable. Twelve participants had a PMData-provided HRmax of undocumented provenance. Three missing values were estimated as 208 - 0.7 × age. Exact session-date resting HR was evaluated separately when available.

$$tHRR\text{-}I(\lambda) = [\Sigma_i P_i c_i \exp(\lambda c_i)] / [\Sigma_i P_i \exp(\lambda c_i)], \lambda \geq 0.$$

$$\text{Continuous } tHRR\text{-}I(\lambda) = [\Sigma_t \Delta t\, h_t \exp(\lambda h_t)] / [\Sigma_t \Delta t \exp(\lambda h_t)].$$

$$\Delta tilt(\lambda) = tHRR\text{-}I(\lambda) - \mu HRR.$$

P_i is the time proportion in HRR bin i, c_i is its midpoint, and h_t is clipped continuous HRR. The ten-bin primary representation used c_i=(i-0.5)/10. The transform is bounded and monotonic in λ. At the full-development λ of 6.2, adjacent deciles differ 1.86-fold in relative weight and HRR=0.95 receives 265-fold the pre-normalization weight of HRR=0.05. Δtilt responds to dispersion and higher moments, including asymmetry and upper-tail exposure. It is not specific to discrete high-intensity events.

Comparator definitions were: linear decile score=$\Sigma_i P_i$ i; mean HRR=$\Sigma_t \Delta t\, h_t/\Sigma_t \Delta t$; entropic HRR=log[$\Sigma_i P_i \exp(\lambda c_i)$]/λ; power-mean HRR=[$\Sigma_i P_i c_i$^q]^(1/q); original exponential score=$\Sigma_i P_i$ exp(k i). Banister TRIMP was integrated from minute-level HRR with sex-specific coefficients and included duration.

S4. Nested evaluation and uncertainty

For the central incremental analysis, the outer loop held out one entire participant. Within each outer training set, λ for Δtilt was searched from 0.0 to 15.0 in steps of 0.1 by inner participant-grouped leave-one-participant-out validation. The inner criterion was participant-balanced MAE for mean HRR plus Δtilt, matching the outer evaluation endpoint. The smallest λ resolved ties. A participant-balanced linear calibration model then predicted the held-out participant, with each training participant contributing equal total weight.

An earlier formula-family sensitivity separately searched λ from 0.0 to 15.0, q from 1.0 to 12.0, and original k from 0.01 to 1.50 using median within-person Spearman correlation among eligible training participants. Tables S3-S5 and S18-S20 retain that exploratory family-selection analysis for transparency; it is not the central incremental comparison in Table S27.

Reported MAE and RMSE are participant-balanced. Pooled R^2 is included as a complementary calibration summary. The 95% intervals use 5000 participant-cluster bootstrap resamples with seed 20260729. They condition on the realized held-out predictions and do not rerun formula-family selection. A separate 200-split analysis with seed 20260731 randomly assigned 12 participants to training and three to test, and repeated λ selection and calibration for the locked tHRR-I family.

Supplementary Table S3. Discretization and resting-HR sensitivity

Estimator	Sessions/participants	Full λ	Outer λ median [IQR]	Balanced MAE	Median within-person rho
5 bins	255/15	12.6	12.6 [12.6 to 13.3]	1.284	0.708
10 bins (development candidate)	255/15	6.2	8.9 [6.2 to 9.3]	1.206	0.666
20 bins	255/15	6.3	7.4 [6.3 to 9.9]	1.223	0.644
Continuous time	255/15	6.1	6.1 [6.1 to 9.1]	1.239	0.644
10 bins + session-date HRrest	209/12	13.0	13.0 [9.2 to 13.0]	1.277	0.548

Parameters are not directly comparable across bin widths because the support points differ. The session-date HRrest analysis used the available subset.

Supplementary Table S4. Complete nested model performance after alternative session matching

Rule	Sessions/ participants	Full λ	tHRR-I MAE	Linear MAE	MAE difference [95% conditional CI]
15-90-min report window	232/15	6.0	1.221	1.309	-0.089 [-0.149 to -0.029]
30-120-min report window	240/15	6.0	1.207	1.290	-0.083 [-0.151 to -0.024]
Duration difference ≤10 min	220/15	7.7	1.212	1.280	-0.068 [-0.130 to -0.009]
Delay-priority cost	264/15	9.3	1.202	1.297	-0.095 [-0.165 to -0.029]
Duration-priority cost	264/15	9.3	1.202	1.297	-0.095 [-0.163 to -0.026]

Each rule rebuilt the eligible session set and repeated participant-level parameter selection and calibration. Intervals remain conditional and exploratory.

Across 200 repeated 12/3 participant splits, median selected λ was 9.3 (IQR 8.9 to 9.3). Median tHRR-I minus linear MAE was -0.094 RPE units (IQR -0.135 to -0.038), and 86.0% of splits favored tHRR-I. These are partition-sensitivity summaries, not external-validation confidence intervals.

S5. Raw-signal and physiological-anchor sensitivities

Supplementary Table S5. Raw-signal and anchor sensitivity of nested prediction

Variant	Sessions/ participants	Full λ	tHRR-I MAE	Linear MAE	MAE difference [95% conditional CI]
Rebuilt primary analysis	255/15	6.2	1.206	1.302	-0.097 [-0.163 to -0.029]
Winsorize upper 0.5% of session HR	255/15	6.2	1.203	1.302	-0.099 [-0.169 to -0.031]
Winsorize upper 1% of session HR	255/15	6.2	1.208	1.302	-0.094 [-0.158 to -0.028]
Cap HRR at 0.95	255/15	6.2	1.206	1.302	-0.097 [-0.161 to -0.030]
Cap sampling gaps at 15 s	255/15	6.2	1.206	1.302	-0.097 [-0.165 to -0.033]
Require Fitbit confidence=3	0/0	Not estimable	Not estimable	Not estimable	Not estimable
HRmax +5 beats·min^{-1}	255/15	6.9	1.226	1.310	-0.083 [-0.146 to -0.019]
HRmax -5 beats·min^{-1}	255/15	6.8	1.208	1.306	-0.098 [-0.155 to -0.040]
HRrest +3 beats·min^{-1}	255/15	5.8	1.206	1.301	-0.096 [-0.162 to -0.029]
HRrest -3 beats·min^{-1}	255/15	6.2	1.215	1.301	-0.086 [-0.152 to -0.023]
Age-predicted HRmax for all	255/15	4.6	1.374	1.406	-0.032 [-0.084 to 0.026]

The rebuilt baseline reproduced archived bin proportions to machine precision. No session met the primary 80% coverage rule when Fitbit confidence=3 alone was required. The all-Tanaka result shows that HRmax specification can materially affect calibration.

S6. Outcome-scale and high-RPE sensitivities

Supplementary Table S6. Exploratory outcome sensitivities

Analysis	Eligible data	tHRR-I	Linear	Difference
Participant-balanced high-RPE AUC	8 participants; 221 sessions; 50 high	0.817	0.673	0.144 [0.028 to 0.268]
Participant-specific quadratic-weighted kappa	9 participants	0.416	0.074	0.181

High RPE was defined post hoc as RPE ≥8. AUC was calculated only for participants who had both classes. Rounded 1-10 predictions were used for quadratic-weighted kappa.

Session-RPE load equals RPE multiplied by duration. The objective load scores also include duration. Their association is therefore partly mathematically coupled and does not provide independent physiological validation.

S7. Clipping, prediction range, and safeguards

Before clipping, 0.034% of weighted valid time was below HRR=0 and 0.649% was above HRR=1. Any below-zero values occurred in 12 sessions and any above-one values in 22 sessions. Exact session-date resting HR was available for 209 sessions.

Supplementary Table S7. Held-out prediction range

Model	Minimum	Maximum	Outside archived 1-10 range
tHRR-I	2.850	7.169	0
Linear decile	3.389	7.100	0
Mean HRR	3.516	7.272	0
Intercept only	5.819	6.141	0

S8. Reproducibility specification

Required Python versions were Python 3.12.13, pandas 2.2.3, NumPy 2.3.5, openpyxl 3.1.5, and python-docx 1.2.0. Fixed random seeds are declared in each analysis script; reviewer_round3_analysis.py used seed 20260730.

Run order: (1) configure the licensed source-data roots; (2) run run_analysis.py, including reviewer_round5_analysis.py; (3) run run_external_analysis.py, including reviewer_round5_weee_agreement.py; (4) build the main and external figures; and (5) build the submission documents. Individual scripts may also be run in the order documented in README.md. Third-party raw data are not redistributed.

All derived CSV and JSON outputs used in the manuscript are archived under 04_Supplement/Derived_results. Source URLs, licensing information, and integrity manifests are archived under 05_Reproducibility. Paths in analysis scripts are relative or configurable; no author-specific path is required.

S9. Claims not resolved by internal reanalysis

The remaining unresolved target is direct external session-RPE validation of the locked formula. None of the screened sources provided an unambiguous session-level link among time-resolved HR, standardized RPE, and measured resting and maximal anchors. The current evidence also lacks blinded hand-labeled PMData linkage and prospective decision-impact outcomes. WEEE and Malaga address independent physiological constructs and signal transportability, but not the direct RPE target. These limits must not be inferred away from the present analyses; a confirmatory study should retain the fixed ten-bin formula and $\lambda=6.2$.

S10. Locked external-analysis protocol

The external protocol was written on 30 July 2026 before inspecting external outcomes. It was not registered and must not be described as preregistered. The PMData development estimate $\lambda=6.2$, the normalized ten-bin formula, the HRR clipping rule, and the primary comparator were frozen. External data could not be used to select another formula family, bin count, λ, or comparator hierarchy. Dataset eligibility was determined from access and variables rather than results.

The external work addresses construct convergence, measurement transportability, and incremental prediction. It does not constitute external session-RPE validation because no screened external dataset provided an unambiguous session-level link between time-resolved HR, defensible HRR anchors, and RPE. Participants were the independent resampling unit. Cross-validation retained all observations from a participant within one fold, and bootstrap intervals resampled participants with replacement.

Supplementary Table S10. External dataset roles

Dataset	Role	Independent outcome	Included observations
Malaga treadmill tests v1.0.1	Primary external construct	180-s excess recovery VO_2	758 tests; 668 participants
WEEE	Secondary construct and signal transportability	Stage VO_2; chest-signal agreement	77 stages; 16 participants

S11. Malaga processing, attrition, and results

The effort phase was identified from treadmill speed greater than 5.05 km·h^{-1}. The lower HR anchor was the minimum stable 30-second rolling median in an identifiable pre-effort segment and was not interpreted as clinical resting HR. HRmax was the highest quality-controlled HR during effort. The primary endpoint was the trapezoidal integral of VO_2 above the minimum smoothed recovery VO_2 during the first 180 seconds. Models used participant-grouped 10-fold validation. The base model contained effort duration, mean HRR, age, sex, and body mass; the augmented model added Δtilt.

Supplementary Table S11. Malaga test-level attrition

Status	Reason	Tests
excluded	lower_anchor_unavailable	2
excluded	pre_effort_shorter_than_60s	68
excluded	recovery_shorter_than_180s	145
excluded	recovery_vo2_unavailable	19
included	included	758

Supplementary Table S12. Malaga participant-balanced external performance

Metric	Base	Base + Δtilt	Augmented minus base (95% CI)
MAE, mL	347.404	347.412	0.009 (-1.234 to 1.276)

Metric	Base	Base + Δtilt	Augmented minus base (95% CI)
RMSE, mL	455.685	456.201	0.516 (-0.621 to 1.757)
R^2	0.2338	0.2321	-0.0017 (-0.0059 to 0.0021)

The full-sample Δtilt coefficient was 6.587 mL per 0.01 HRR unit (95% participant-cluster CI -6.279 to 19.527). This coefficient was descriptive and did not demonstrate incremental out-of-sample prediction.

Supplementary Table S13. Malaga exploratory rank associations

Outcome	Association	ρ or Δρ	95% participant-cluster CI
180-s excess recovery VO2	mean_hrr	-0.049	-0.122 to 0.025
180-s excess recovery VO2	thrr_i	-0.015	-0.087 to 0.059
180-s excess recovery VO2	delta_tilt	0.069	-0.006 to 0.142
180-s excess recovery VO2	thrr_i_minus_mean_hrr	0.034	-0.031 to 0.100
peak exercise VO2	mean_hrr	-0.076	-0.154 to 0.000
peak exercise VO2	thrr_i	-0.052	-0.127 to 0.022
peak exercise VO2	delta_tilt	0.064	-0.012 to 0.142
peak exercise VO2	thrr_i_minus_mean_hrr	0.023	-0.045 to 0.093
one-minute heart-rate recovery	mean_hrr	-0.141	-0.217 to -0.064
one-minute heart-rate recovery	thrr_i	-0.050	-0.126 to 0.028
one-minute heart-rate recovery	delta_tilt	0.138	0.063 to 0.216

Outcome	Association	ρ or Δρ	95% participant-cluster CI
one-minute heart-rate recovery	thrr_i_minus_mean_hrr	0.091	0.026 to 0.160
peak exercise RER	mean_hrr	0.195	0.121 to 0.267
peak exercise RER	thrr_i	0.122	0.045 to 0.195
peak exercise RER	delta_tilt	-0.194	-0.265 to -0.121
peak exercise RER	thrr_i_minus_mean_hrr	-0.073	-0.134 to -0.013
peak recovery RER	mean_hrr	0.178	0.103 to 0.250
peak recovery RER	thrr_i	0.102	0.030 to 0.173
peak recovery RER	delta_tilt	-0.171	-0.244 to -0.099
peak recovery RER	thrr_i_minus_mean_hrr	-0.076	-0.137 to -0.013

The Malaga secondary intervals used 1,000 participant-cluster bootstrap replicates for computational feasibility and are explicitly exploratory. Their mixed directions were not used to redefine the primary endpoint or claim.

S12. WEEE processing and results

Six five-minute WEEE stages were reconstructed: sitting, standing, low- and high-intensity cycling, and low- and high-intensity running. The source contained 102 possible stages from 17 participants. P10 lacked a sitting HR anchor, removing six stages. Nineteen of the remaining 96 stages failed reference HR or VO_2 coverage, leaving 77. Attrition was intensity dependent, with only 3 of 16 high-running stages retained.

Supplementary Table S14. WEEE exploratory convergence with stage VO_2

Index	Spearman ρ	95% participant-cluster CI
Mean HRR	0.757	0.614 to 0.870
tHRR-I	0.782	0.653 to 0.878
Δtilt	0.457	0.234 to 0.638
tHRR-I minus mean HRR	0.025	-0.003 to 0.058

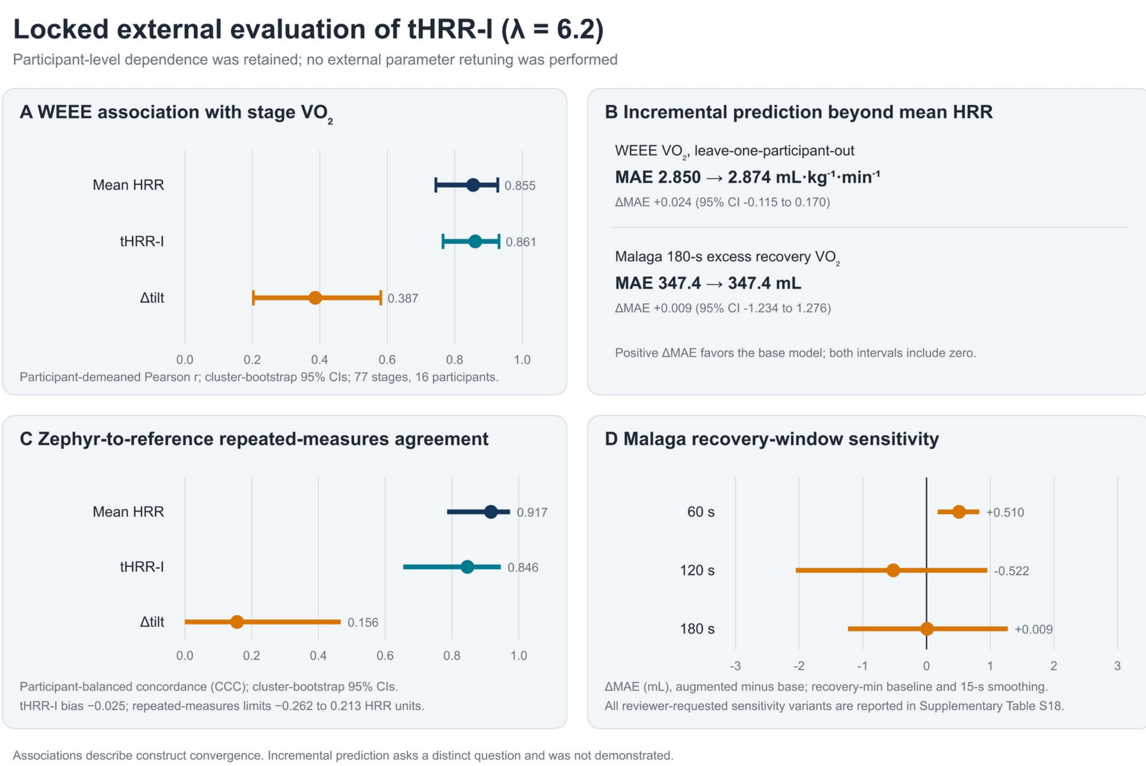
Supplementary Table S15. WEEE participant-balanced leave-one-participant-out VO_2 performance

Metric	Base	Base + Δtilt	Augmented minus base (95% CI)
MAE, $mL \cdot kg^{-1} \cdot min^{-1}$	2.850	2.874	0.024 (-0.115 to 0.170)
RMSE, $mL \cdot kg^{-1} \cdot min^{-1}$	3.897	3.936	0.039 (-0.074 to 0.147)
R^2	0.6673	0.6606	-0.0067 (-0.0244 to 0.0136)

Supplementary Table S16. Zephyr versus WEEE reference chest-strap agreement

Index	Bias	MAE	RMSE	ICC(A,1)
mean_hrr	-0.026	0.045	0.081	0.920
thrr_i	-0.026	0.063	0.120	0.854
delta_tilt	0.000	0.030	0.057	0.179

Apple Watch agreement was based on one participant and two aligned stages, and Empatica E4 agreement on two participants and four stages. These estimates were retained in the machine-readable outputs but were not interpreted because participant coverage was inadequate.



Supplementary Figure S1/Main Figure 5. External construct and measurement transportability summary. λ=6.2 was fixed from PMData; no external retuning was performed.

S13. Dataset screening and reporting safeguards

Supplementary Table S17. External dataset screening

Dataset	Decision	Reason
University of Malaga treadmill maximal exercise tests	Included	Meets the locked construct-validation criteria
WEEE	Included	Meets secondary construct and device-transportability criteria
SoccerMon	Excluded after feasibility audit	Objective archives total 99.1 GB and the available documentation did not establish an analysis-ready session-level HR anchor and RPE linkage within the resources used for this revision
Polar futsal fatigue-monitoring record	Excluded after repository audit	Zenodo API record contained zero downloadable files on 2026-07-30

The analysis preserves both favorable and null results. The WEEE convergence association and Zephyr agreement support transportability of the score as a descriptive index. The WEEE and Malaga incremental-prediction contrasts did not support an additional predictive contribution from Δtilt beyond mean HRR for their selected physiological endpoints. These statements do not conflict: convergence asks whether the score tracks graded exposure, whereas incremental prediction asks whether it adds information after mean HRR and covariates are already known.

S14. External reproducibility assets

The external_validation folder contains the locked protocol, dataset-specific implementation decisions, source-file manifests, SHA-256 digests, selected WEEE extraction records, quality-control tables, derived test- and stage-level data, fixed-seed analysis scripts, and machine-readable summaries. The Malaga raw files were verified against PhysioNet checksums. WEEE files were selectively extracted from the public ZIP by HTTP range requests and verified by ZIP CRC and SHA-256. Third-party raw files should not be uploaded to the public code repository without a separate redistribution review.

S15. Selection-aware development uncertainty and matching confidence

The selection-aware outer leave-one-participant-out analysis reselected the transformation family and its parameter using only each training fold. It then fitted participant-balanced calibration and evaluated the selected pipeline in the held-out participant. This procedure covers selection among the retrospectively defined candidate families, but it cannot quantify the earlier choice to invent those families after inspecting PMData.

Supplementary Table S18. PMData selection-aware and strict-matching analyses

Analysis	Sessions/ participants	Selected model	Model MAE	Linear MAE	Difference (95% CI)
Selection-aware outer LOPO	255/15	tHRR-I in 14 folds; power mean in 1	1.208	1.302	-0.094 (-0.161 to -0.031)
Strict matching tier	203/15	tHRR-I	1.210	1.274	-0.064 (-0.123 to -0.013)

All 255 primary matches were bidirectionally unique and had one eligible report and one exercise candidate under the primary rule. The strict tier required a 15-120-minute reporting delay and a duration difference of at most 10 minutes. Its median delay was 60.4 minutes, and its median duration difference was 3.1 minutes.

Supplementary Table S19. Circular within-participant RPE label-permutation negative control

Quantity	Observed	Selection-adjusted null
Selected λ	6.2	Reselected in every replicate
Median within-participant Spearman correlation	0.704	95th percentile of null maximum: 0.193
Permutation replicates	5,000	Circular shifts retained each participant's marginal RPE distribution
Empirical P value	0.00020	(1 + exceedances)/(1 + replicates)

Supplementary Table S20. Exploratory full-sample λ heterogeneity

Stratum	Sessions	Eligible participants	Selected λ	Interpretation
All	255	9	6.2	Development estimate
Treadmill	143	7	7.4	Exploratory
Running	108	7	1.9	Exploratory

Male	215	7	13.2	Exploratory
Female	40	2	7.5	Too sparse for inference
Training status	NA	NA	Not estimable	Variable unavailable

S16. Expanded WEEE repeated-measures and agreement analyses

WEEE stages are repeated measurements nested within participants. Participant-centered correlations quantify within-participant stage change, and two-way residual correlations remove participant and stage means. Primary Zephyr agreement now uses equal total weight per participant. Bias is the mean of participant-specific biases. Repeated-measures limits combine between- and within-participant difference variance. Participant-balanced concordance is reported with participant-cluster bootstrap intervals. The original stage-level ICC(A,1) remains a secondary descriptor.

Supplementary Table S21. WEEE repeated-measures associations with VO_2

Association level	Mean HRR r	tHRR-I r	Δtilt r	tHRR-I minus mean HRR
Within participant	0.855 (0.744 to 0.928)	0.861 (0.766 to 0.931)	0.387 (0.203 to 0.581)	0.006 (-0.008 to 0.027)
Between participant	0.657 (0.371 to 0.863)	0.643 (0.357 to 0.852)	0.310 (-0.138 to 0.687)	-0.015 (-0.067 to 0.035)
Participant + stage residual	0.367 (-0.023 to 0.678)	0.362 (0.002 to 0.662)	0.061 (-0.101 to 0.218)	-0.005 (-0.051 to 0.042)

Supplementary Table S22. Zephyr versus reference chest-signal agreement with uncertainty

Index	Bias (95% CI)	MAE	ICC(A,1) (95% CI)	Bland-Altman limits
Mean HRR	-0.026	0.045	0.920 (0.791 to 0.974)	-0.177 to 0.124
tHRR-I	-0.026 (-0.046 to -0.001)	0.063	0.854 (0.667 to 0.952)	-0.257 to 0.205
Δtilt	0.000	0.030	0.179 (-0.013 to 0.493)	-0.113 to 0.113

Supplementary Table S23. WEEE timestamp synchronization audit

Signal	Stages	Participants	Median coverage within 2 s	Median absolute offset	Median HR cross-correlation lag
Zephyr	77	16	100%	0.429 s	+6 s
Apple Watch	2	1	Sparse	Reported in CSV	Not interpreted
Empatica E4	4	2	Sparse	Reported in CSV	Not interpreted

Absolute repository timestamps were used without manual lag correction. Subsecond sample matching indicates adequate clock alignment for Zephyr. The +6-second cross-correlation lag may reflect sensor processing, smoothing, or physiological response differences and is retained as a measurement limitation.

S17. Malaga endpoint and anchor sensitivity

The primary Malaga variable is an excess recovery VO_2 integral relative to the minimum of the analyzed recovery window. It is not conventional EPOC because the archive did not provide a uniform preexercise resting VO_2 baseline. The analyses below vary the recovery window, endpoint baseline, smoothing span, and HR lower anchor. No variant produced a stable incremental benefit for Δtilt.

Supplementary Table S24. Malaga recovery-endpoint sensitivity

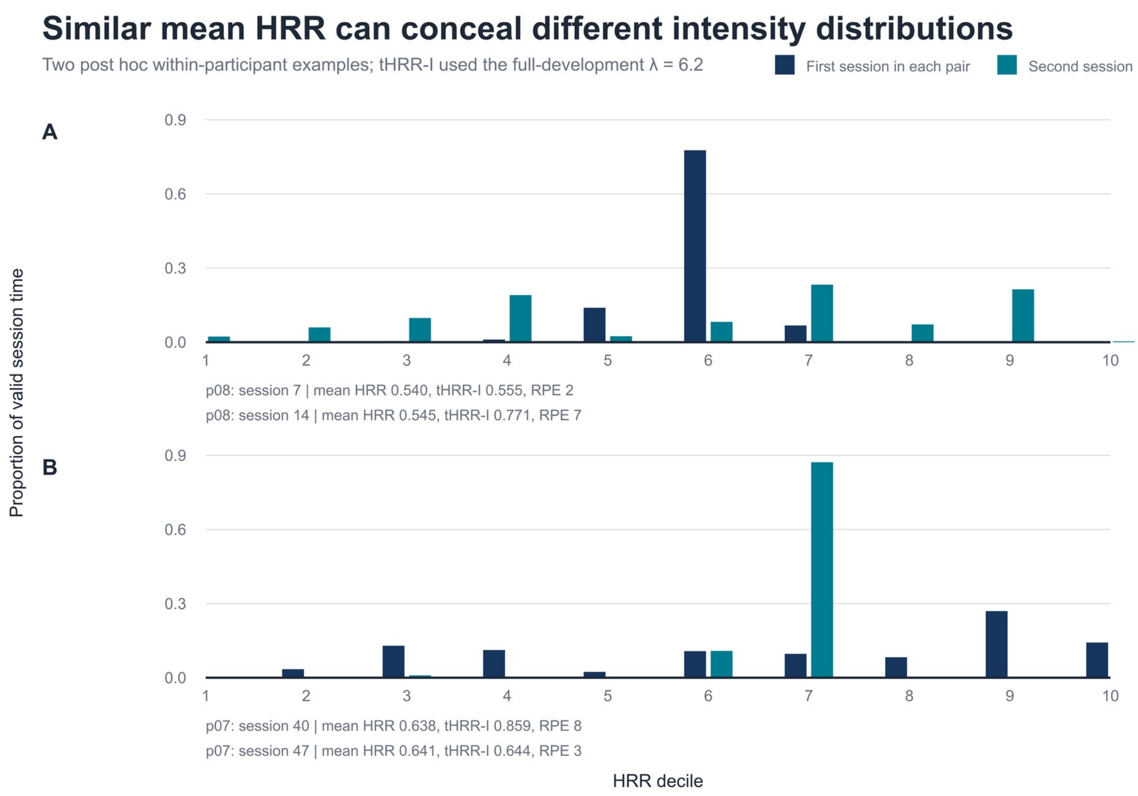
Window/definition	Tests/participants	Base MAE	Base + Δtilt MAE	ΔMAE (95% CI)
60 s, recovery minimum	757/667	131.854	132.364	0.510 (0.174 to 0.828)
60 s, final 30 s	757/667	84.015	84.368	0.353 (-0.217 to 0.902)
60 s, pre-effort baseline	753/665	261.351	261.874	0.523 (-1.344 to 2.469)
120 s, recovery minimum	757/667	272.893	272.371	-0.522 (-2.053 to 0.953)
120 s, final 30 s	757/667	253.055	253.196	0.141 (-0.777 to 1.125)
120 s, pre-effort baseline	753/665	405.723	404.952	-0.771 (-4.029 to 2.569)
180 s, recovery minimum (primary)	758/668	347.404	347.412	0.009 (-1.234 to 1.276)
180 s, final 30 s	758/668	320.580	320.927	0.346 (-1.197 to 1.937)
180 s, pre-effort baseline	754/666	510.566	510.256	-0.310 (-4.517 to 3.921)

Supplementary Table S25. Malaga HR lower-anchor sensitivity for the primary endpoint

HR lower anchor	Base MAE	Base + Δtilt MAE	ΔMAE (95% CI)
Minimum rolling 30-s pre-effort median	347.404	347.412	0.009 (-1.234 to 1.276)
Median of first 60 s	347.062	347.105	0.043 (-1.570 to 1.707)
Median of full pre-effort segment	346.990	347.095	0.106 (-1.037 to 1.305)

S18. Post hoc illustrations and implementation concept

Supplementary Figure S2 gives two exhaustively selected, post hoc within-participant examples. Each pair has nearly equal mean HRR but a different HRR distribution. The examples explain the transform and are not validation evidence.



Supplementary Figure S2. Post hoc equal-mean examples. Bars show valid-time proportions in HRR bins for the two highest-ranked within-participant pairs from different participants. Values use λ=6.2. The pairs were selected for illustration and are not inferential evidence.

Supplementary Table S26. Proposed retrospective review display, pending prospective validation

Display	Review question	Guardrail
Mean HRR	Was typical cardiovascular intensity consistent with the plan?	Compare within athlete and verify anchors.
Δtilt	Was upper-tail exposure greater than in a similar-mean session?	No universal cutoff; inspect context and HR trace.
Duration and duration × Δtilt	Did tail-sensitive exposure accumulate over time?	Interpret with RPE, external load, and recovery.
Quality/context flags	Could coverage, mode, heat, stress, or sensor fit explain the value?	Do not automate decisions when quality is uncertain.

S19. Incremental validity, flow reconciliation, and repeated-measures agreement

The reviewer-round-5 candidate set was fixed before the final outer-fold readout. No candidate was deleted because it underperformed. The central comparison asks whether Δtilt adds predictive information after binned mean HRR is already in the model. λ selection uses participant-balanced MAE in inner participant-grouped leave-one-participant-out validation. Fixed λ=6.2 rows are development-condition sensitivities and are not independent validation.

Supplementary Table S27. PMData incremental and transparent-baseline comparison

Model	MAE	ΔMAE vs mean HRR [95% interval]	Favored, n/15	Exact P
Mean HRR	1.302	Reference	–	–
Mean HRR + Δtilt, nested λ	1.232	-0.070 [-0.165 to 0.035]	11	0.228
Mean HRR + Δtilt, fixed λ=6.2	1.228	-0.074 [-0.155 to 0.002]	10	0.098
tHRR-I alone, fixed λ=6.2	1.228	-0.074 [-0.133 to -0.017]	9	0.030
Mean HRR + variance	1.294	-0.008 [-0.064 to 0.054]	9	0.798
Mean HRR + time ≥80% HRR	1.279	-0.023 [-0.136 to 0.107]	9	0.721
Selection-aware transparent pipeline	1.245	-0.057 [-0.151 to 0.048]	11	0.312

Negative ΔMAE favors the candidate. Bootstrap intervals resample the 15 participant-level paired losses. Exact P values enumerate all 2^{15} sign flips. The fully nested Δtilt model reduced average MAE for 11 participants, but its interval included zero.

Supplementary Table S28. PMData sample-flow reconciliation

Analysis stage or rule	Selected/broad pairs	Unique pairs	Original primary IDs retained	Rebuilt analysis sessions
Broad primary construction	469	447	–	255 (267 passed HR QC before all criteria)
Primary 15-180 min	449	443	255	255
15-90 min	401	398	225	232
30-120 min	413	408	232	240
Duration difference ≤10 min	363	357	214	220
Delay-priority matching cost	449	443	255	264
Duration-priority matching cost	449	443	255	264

The broad construction counts describe sequential attrition from the archived dataset. Sensitivity rules rebuild the candidate graph. Therefore, rebuilt samples can contain newly eligible pairs that were not among the 255 original primary IDs.

Supplementary Table S29. WEEE participant-stage retention

Stage	Eligible after anchor QC	Included	Excluded	Retention
Sitting	16	16	0	100.0%
Standing	16	16	0	100.0%
Low cycling	16	15	1	93.8%
High cycling	16	14	2	87.5%
Low running	16	13	3	81.3%
High running	16	3	13	18.8%

Supplementary Table S30. Zephyr repeated-measures agreement

Index	Participant-balanced bias	Participant-balanced CCC [95% CI]	Repeated-measures limits	Participant-centered r
Mean HRR	-0.025	0.917 [0.786 to 0.974]	-0.182 to 0.132	0.925
tHRR-I	-0.025	0.846 [0.654 to 0.946]	-0.262 to 0.213	0.846
Δtilt	0.000	0.156 [-0.007 to 0.467]	-0.113 to 0.113	0.171

All Zephyr estimates use 77 paired stages from 16 participants. Concordance and bias give each participant equal total weight. Limits of agreement use the sum of between- and within-participant difference variance components. Confidence intervals resample participants.